Patellar Tendon Load Progression during Rehabilitation Exercises: Implications for the Treatment of Patellar Tendon Injuries

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ABSTRACT

SCATTONE SILVA, R., K. SONG, T. J. HULLFISH, A. SPRAGUE, K. G. SILBERNAGEL, and J. R. BAXTER. Patellar Tendon Load Progression during Rehabilitation Exercises: Implications for the Treatment of Patellar Tendon Injuries. *Med. Sci. Sports Exerc.*, Vol. 56, No. 3, pp. 545–552, 2024. **Purpose:** This study aimed to evaluate patellar tendon loading profiles (loading index, based on loading peak, loading impulse, and loading rate) of rehabilitation exercises to develop clinical guidelines to incrementally increase the rate and magnitude of patellar tendon loading during rehabilitation. **Methods:** Twenty healthy adults (10 females/10 males, 25.9 ± 5.7 yr) performed 35 rehabilitation exercises, including different variations of squats, lunge, jumps, hops, landings, running, and sports specific tasks. Kinematic and kinetic data were collected, and a patellar tendon loading index was determined for each exercise using a weighted sum of loading peak, loading rate, and cumulative loading impulse. Then the exercises were ranked, according to the loading index, into tier 1 (loading index ≤0.33), tier 2 (0.33 < loading index <0.66), and tier 3 (loading index ≥0.66). **Results:** The single-leg decline squat showed the highest loading index (0.747). Other tier 3 exercises included single-leg forward hop (0.666), single-leg countermovement jump (0.711), and running cut (0.725). The Spanish squat was categorized as a tier 2 exercise (0.563), as was running (0.612), double-leg countermovement jump (0.610), single-leg drop vertical jump (0.599), single-leg full squat (0.580), double-leg drop vertical jump (0.563), lunge (0.471), double-leg full squat (0.428), single-leg 60° squat (0.411), and Bulgarian squat (0.406). Tier 1 exercises included 20 cm step up (0.187), 20 cm step down (0.288), 30 cm step up (0.321), and double-leg 60° squat (0.224). **Conclusions:** Three patellar tendon loading tiers were established based on a combination of loading peak, loading impulse, and loading rate. Clinicians may use these loading tiers as a guide to progressively increase patellar tendon loading during the rehabilitation of patients with patellar tendon disorders and after anterior cruciate ligament reconstruction using the bone–patellar tendon–bone graft. **Key Words:** PATELLAR TENDINOPATHY, JUMPER'S KNEE, PHYSICAL THERAPY, BIOMECHANICS, ANTERIOR CRUCIATE LIGAMENT, MOTION

Patellar tendinopathy is one of the most common causes of knee pain in young active individuals (1,2). The prevalence of patellar tendinopathy has been reported to be 33% among young athletes (2,3). Among elite volleyball athletes, the prevalence can be as high as 45% (1). Because patellar tendinopathy is a common reason for young individuals to stop being physically active and abandon sports participation (4), if not properly managed, this condition has long-term detrimental health effects. Patellar tendon pain is also a common complication after anterior cruciate ligament (ACL) reconstruction using the bone–patellar tendon–bone graft (5,6). It is possible that this patellar tendon morbidity is related to overly conservative or excessively aggressive rehabilitation programs that fail to gradually progress the quadriceps muscle/patellar tendon load tolerance after surgery. In this context, a more comprehensive knowledge of how much load the patellar tendon undergoes during common rehabilitation exercises is crucial for progressively increasing load in the patellar tendon during the rehabilitation of patellar tendinopathy and after ACL reconstruction.

Exercise is the main intervention for the treatment of patients with patellar tendon injury. The single-leg decline squat is the most commonly prescribed exercise for the rehabilitation of patellar tendinopathy (7), probably because of the fact that the large majority of studies involving patellar tendinopathy patients have used the eccentric single-leg

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decline squat as the main exercise intervention (8–14). However, there is no convincing clinical evidence to demonstrate that isolated eccentric exercise improves clinical outcomes more than other loading therapies (15–17). In fact, similar improvements in pain and function in patellar tendinopathy patients occur with progressive loading protocols regardless of contraction type (11,14,15,18,19). The patellar tendon force has been quantified during the single-leg decline squat (20–22). A few studies have also evaluated the patellar tendon force during activities such as the drop jump (23,24) and the stop jump (25–27). However, little is known about how much load the patellar tendon undergoes during other rehabilitation exercises, such as the lunge, variations of the squat, running, different jumps, and landings.

Recently, Baxter et al. (28) developed a loading index system to characterize loading progression in the Achilles tendon during 25 rehabilitation exercises. A loading index for each exercise was established based on a comprehensive dataset that factored not only loading peak but also loading rate and the cumulative loading impulse (28). To our knowledge, such a complete loading dataset is still missing for exercises typically used for the rehabilitation of patellar tendon injury. In addition, no studies have reported how patellar tendon loading during rehabilitation exercises compares to activities such as walking and running in the same individuals. Patients with an ACL reconstruction using the bone–patellar tendon–bone autograft have been suggested to have a surgically induced tendinopathy at the graft harvest site that should be treated with a progressive loading program (29). However, in the absence of objective data, clinicians need to rely solely on their experience and on theoretical premises when developing rehabilitation exercise protocols for patients, which highlights the importance of quantitative evidence to refine patellar tendon disorders rehabilitation.

In this context, in both treatment of patellar tendinopathy and prevention of patellar tendon pain after ACL reconstruction using the bone–patellar tendon–bone graft, clinicians would benefit from a greater understanding of the loads that act on the tendon during exercises. The purpose of this study was to determine a loading index to quantify, compare, rank, and categorize overall loading levels in the patellar tendon across 35 exercises typically used in the clinical rehabilitation of patients with knee disorders. This loading rank will provide clinicians with meaningful guidelines to progressively load the patellar tendon and improve tissue healing and functional recovery of young active individuals.

METHODS

Participants. Twenty healthy adults were recruited (10 females, 25.9 ± 5.7 yr, 71.2 ± 14.0 kg, 1.71 ± 0.09 m, and 24.1 ± 2.6 $kg \cdot m^{-2}$) from the University campus and local community between January and June 2022. To be included, participants had to be between 18 and 40 yr of age and have no history of injury in the lower limbs or spine in the last 6 months. An experienced physical therapist (RSS) confirmed that each subject had no current ongoing musculoskeletal injuries or history of patellar tendon pain. We decided to study individuals without patellar tendon pain to characterize the loading biomechanics during pain-free movement. Participants provided written informed consent before initiating the procedures, and the study was approved by the institutional review board at the University of Pennsylvania.

Initial procedures. Before data collections, participants changed into standard exercise clothing (running shorts and tank tops) and were provided with a pair of the same running shoes (Air Pegasus; Nike, Beaverton, OR). Thirty-one retroreflective markers were placed over anatomic landmarks of the pelvis, thighs, shanks, feet (on the shoes), and torso (30). On the lower extremity, single markers were placed bilaterally on the anterior thighs, lateral shanks, and medial and lateral femoral epicondyles and malleoli. Markers were also placed over the shoes over the first and fifth metatarsal heads, the distal phalanx of the hallux, and behind the calcaneus.

A static trial was recorded with the participants in the anatomical position to align them with the global coordinates and identify joint axes. Then the medial knee and ankle markers were removed so they would not interfere with each participant's natural movement during the evaluated exercises. A 5-min walk on a treadmill at a self-selected pace was used as warm-up.

Data collection. For the biomechanical data collection, a 12-camera motion capture system was used at sampling frequency of 100 Hz (Raptor Series, Motion Analysis Corp., Rohnert Park, CA). Ground reaction forces were simultaneously collected using three force platforms embedded under the floor at a frequency of 1000 Hz (BP600900; AMTI, Watertown, MA).

Participants performed 35 exercises that are typically used in the rehabilitation of patellar tendinopathy and after ACL reconstruction (8,11,14,29,31–36). The exercises included different variations of squats (half squat, full squat, sumo squat, Bulgarian squat, Spanish squat, single-leg decline squat), step up and step down, lunge, countermovement jumps and hops (double and single-legged), drop landings, drop vertical jumps and hops, forward jumps and hops, repetitive forward and lateral jumps and hops, running, run-and-cut, running deceleration (run-and-stop), split jumps, and sports specific jumps. A detailed description of the exercises is presented in Supplemental Table 1 (Supplemental Digital Content 1, Demonstration of the 35 exercises assessed in the study, http://links.lww.com/MSS/C926). The exercises were divided into four modules, in an order that was determined during pilot testing to minimize the effects of physical exhaustion. Participants rested at least 1 min between each exercise while the set up for the next exercise was prepared and while a physical therapist demonstrated the next exercise. Between modules, participants were provided 3–5 min of rest periods (37). Participants were provided visual demonstrations and verbal guidance throughout each exercise by the same physical therapist with more than 10 yr of clinical experience. All participants performed the exercises in the same order. At least three successful repetitions of each exercise were collected, including three

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back-and-forth repetitions during exercises such as the split jumps. Repetitions were considered successful if the exercise was completed at the force plate without losing balance.

Data analysis. Marker trajectories were labeled, and the data were imported into a biomechanical model (Visual3D; C-Motion, Germantown, MD). A constrained link-segment model was used for best estimates of segment positions (30). This model constrains the knee joint to sagittal plane motion only, while allowing the hip and ankle joints to have all 3 degrees of freedom. Slater et al. (30) have shown that this model can accurately track segment positions with the simple marker set that was used. The individual exercise repetitions within each trial were extracted using the ground reaction forces or manually defined start and stop events for the subsequent biomechanical analysis. Each exercise was considered in its entirety for data analysis, i.e., both the eccentric and the concentric phases of squats and both the weight acceptance and the propulsion phases of jumps. For all but one exercise, ground reaction forces represented all external forces applied onto the participants. The only exception was the Spanish squat, which uses a belt placed behind the proximal shank to support the subjects while they squat with their center of body mass behind the heels. Because the Spanish squat is quasi-static, the belt forces applied to the proximal shanks were estimated by assuming that the force through either belt strap is of equal magnitude and opposite direction to the horizontal ground reaction force under the corresponding foot. These belt forces were assigned at 20% of the shank length from the knee joint toward the ankle joint in the model to match the experimental setup.

To quantify patellar tendon loading during each exercise, the knee joint flexion angles and extension moments from the link-segment model were calculated using inverse kinematics and inverse dynamics algorithms (Visual3D, C-Motion) (38). The right lower limb was chosen as the side of interest for each participant to standardize testing. It was also the dominant lower limb of all participants. The data were filtered using a fourth-order Butterworth low-pass filter with a 6-Hz cutoff frequency (38).

For determining patellar tendon force, first, a knee angle-dependent cubic polynomial function for patellar tendon moment arms (equation 1) was derived from a recent imaging study database, which reported data from 0° (neutral) to 120° of knee flexion (39). This polynomial was extrapolated for patellar tendon moment arms during hyperextension up to 10° and set moments arms in extreme knee flexion to equal of 120° when it approached a plateau (39,40). Next, the patellar tendon force was estimated by dividing the knee moment by the patellar tendon moment arm at the specific knee angle during each time instance of motion (equation 2).

$$\mathrm{MA_{PT}}(\theta_t) = 4.66e^{-5}\theta_t^{\,3} - 9.43e^{-3}\theta_t^{\,2} + 0.397\theta_t + 46.7\ (\mathrm{unit:mm}) \qquad [1]$$

$$F_{\mathrm{PT}}(\theta_t) = M_{\mathrm{knee}}(\theta_t)/\mathrm{MA_{PT}}(\theta_t)\ (\mathrm{unit:N}), \qquad [2]$$

where MA is moment arm, PT is patellar tendon, F is force, θ is knee flexion angle, and t is time.

Patellar tendon force was normalized by the participant's body weight, and loading peak, loading impulse, and loading rate were determined following procedures adapted from Baxter et al. (28). Briefly, loading peak was calculated as the maximum force in each repetition; loading impulse was determined as the area under the force–time curve over each repetition; and loading rate was calculated as the maximum instantaneous change of force overtime (Fig. 1) (28). Average values of each exercise repetition were used for analysis, including pairs of alternating motions where the right leg was in a leading/trailing or abducted/adducted positions.

For the group analysis, intersubject averages and standard deviations were calculated for each metric of the patellar tendon force (peak, impulse, rate) during all 35 exercises. To quantify the overall patellar tendon load during each exercise, the group average of the loading metrics was used to compute a weighted loading index (28). For this, loading peak, impulse, and rate of the patellar tendon force were normalized by its maximum among the 35 exercises. Then a weighted sum was computed with 50% of weight on loading peak, 30% on impulse, and 20% on rate, resulting in a single loading index for each exercise (28). The loading index ranges from 0 to 1 theoretically, with 0 representing no load and 1 representing an exercise that would have loading peak, impulse, and rate all reaching maximum. For comparison of the overall loading levels among exercises, the loading index of all 35 exercises was ranked in an ascending order (from 0 to 1), and the index was categorized into 3 tiers: tier 1 (low, ≤0.33), tier 2 (moderate, 0.33–0.66), and tier 3 (high, ≥0.66).

Data sharing. The biomechanical data and the musculoskeletal models that were used are available on a public repository (https://zenodo.org/records/10076568). A modifiable worksheet that embeds the group average loading metrics is also presented as Supplemental Digital Content (see Supplemental Digital Content 2, http://links.lww.com/MSS/C927), so anyone can rerank and recategorize the loading index based on user-selected loading metric weights.

RESULTS

The patellar tendon loading index ranged from 0.106 during walking to 0.747 during the single-leg decline squat (Fig. 2). Tier 1 exercises (loading index ≤0.33) included walking, low step up, 60° double-leg squat, low step down, and high step up. These exercises involved peak loads ranging from 0.8 to 2.7 body weights (Table 1). Tier 2 exercises (0.33 < loading index <0.66) included high step down, lunge, running, running deceleration, and most squats and jump–landing exercises. The peak loads of the tier 2 exercises ranged from 2.8 to 5.0 body weights. Tier 3 exercises (loading index ≥0.66) included the single-leg decline squat, run-and-cut, single-leg countermovement hop, single-leg repeated forward hops, and single-leg maximal forward hop. Peak loading in tier 3 exercises ranged from 4.6 to 5.5 body weights (Table 1).

The single-leg decline squat was the exercise with the highest loading index. The squat exercises loaded the patellar tendon differently based on the movement variation and

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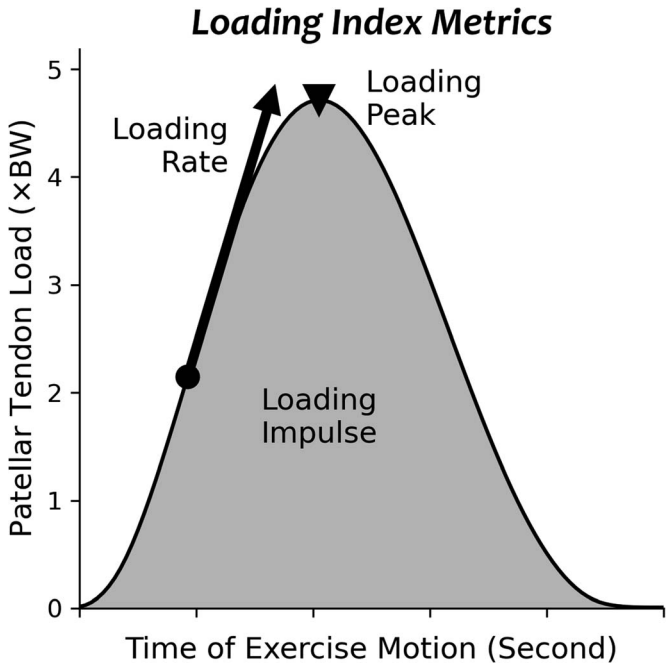

FIGURE 1—Loading peak (maximum force), loading rate (maximum instantaneous change of force overtime), and loading impulse (area under the force–time curve) during a rehabilitation exercise.

whether one or two legs were used to complete the exercise. Figure 3 presents the patellar tendon loading during different variations of squats as well as during walking and running. The single-leg decline squat was the only squat variation that generated more peak patellar tendon loading than running. Single-leg squats did not double peak loading compared with double-leg squats. However, 60° squats resulted in approximately half the peak loading when compared with full squats (Fig. 3).

DISCUSSION

In this study, through the biomechanical analysis of 35 rehabilitation exercises, an exercise order that gradually increases patellar tendon loading was established. To the best of our

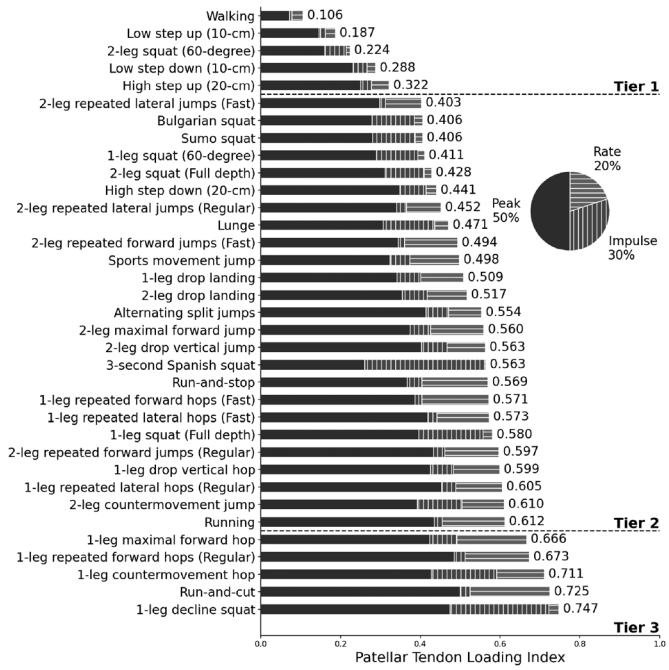

FIGURE 2—Patellar tendon loading index across 35 rehabilitation exercises, categorized as low (tier 1), moderate (tier 2), and high (tier 3). Each loading index was determined with a 50% weight on loading peak (body weight), 30% weight on loading impulse (body weight·second), and 20% on loading rate (body weight/second).

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TABLE 1. Patellar tendon loading index, loading peak, loading impulse, and loading rate across 35 rehabilitation exercises (n = 20).

	Exercise	Loading Index[a]	Loading Peak (BW)	Loading Impulse (BW·s)	Loading Rate (BW/s)
Tier 1	Walking	0.106	0.8 ± 0.4	0.2 ± 0.1	13.4 ± 4.9
	Low step up	0.187	1.6 ± 0.3	0.4 ± 0.1	12.6 ± 2.9
	2-Leg squat (60°)	0.224	1.8 ± 0.5	1.5 ± 0.5	4.7 ± 1.8
	Low step down	0.288	2.5 ± 0.5	0.9 ± 0.4	10.5 ± 2.5
	High step up	0.321	2.7 ± 0.4	0.7 ± 0.2	21.4 ± 3.5
Tier 2	2-Leg repeated lateral jumps (fast)	0.403	3.3 ± 1.0	0.4 ± 0.2	44.0 ± 12.1
	Bulgarian squat	0.406	3.0 ± 0.5	2.9 ± 0.7	10.1 ± 2.2
	Sumo squat	0.406	3.1 ± 0.5	2.9 ± 0.7	9.1 ± 1.7
	1-Leg squat (60°)	0.411	3.2 ± 0.7	2.9 ± 0.7	8.1 ± 2.6
	2-Leg squat (full)	0.428	3.4 ± 0.7	2.7 ± 0.5	9.2 ± 2.7
	High step down	0.441	3.8 ± 0.5	1.8 ± 0.8	12.9 ± 3.1
	2-Leg repeated lateral jumps (regular)	0.452	3.7 ± 0.9	0.7 ± 0.2	42.2 ± 7.5
	Lunge	0.471	3.3 ± 0.6	3.6 ± 0.9	16.8 ± 4.7
	2-Leg repeated forward jumps (fast)	0.494	3.8 ± 1.3	0.4 ± 0.2	64.4 ± 20.3
	Sport movement jump	0.498	3.5 ± 0.7	1.4 ± 0.6	60.3 ± 15.4
	1-Leg drop landing	0.509	3.7 ± 0.8	1.7 ± 0.7	51.4 ± 9.3
	2-Leg drop landing	0.517	3.9 ± 0.8	1.7 ± 0.7	48.1 ± 9.5
	Alternating split jumps	0.554	4.5 ± 0.9	1.5 ± 0.4	40.1 ± 9.5
	2-Leg maximal forward jump	0.560	4.1 ± 0.5	1.4 ± 0.5	64.4 ± 10.4
	2-Leg drop vertical jump	0.563	4.4 ± 0.9	1.8 ± 0.5	46.1 ± 9.0
	3-s Spanish squat	0.563	2.8 ± 1.1	8.2 ± 3.0	1.6 ± 1.3
	Run-and-stop	0.569	4.0 ± 0.9	1.0 ± 0.4	79.6 ± 20.7
	1-Leg repeated forward hops (fast)	0.571	4.2 ± 1.4	0.5 ± 0.2	80.3 ± 17.8
	1-Leg repeated lateral hops (regular)	0.573	4.6 ± 1.4	0.6 ± 0.2	63.2 ± 14.8
	1-Leg squat (full)	0.580	4.3 ± 0.7	4.4 ± 0.9	11.3 ± 2.4
	2-Leg repeated forward jumps (regular)	0.597	4.7 ± 0.8	0.8 ± 0.2	65.1 ± 13.9
	1-Leg drop vertical hop	0.599	4.7 ± 1.1	1.6 ± 0.4	55.8 ± 10.9
	1-Leg repeated lateral hops (regular)	0.605	5.0 ± 1.1	1.0 ± 0.3	56.0 ± 9.5
	2-Leg countermovement jump	0.610	4.3 ± 0.7	3.1 ± 1.0	50.7 ± 10.5
	Running	0.612	4.8 ± 0.8	0.5 ± 0.1	75.7 ± 11.4
Tier 3	1-Leg maximal forward hop	0.666	4.6 ± 0.8	1.9 ± 0.9	83.9 ± 16.2
	1-Leg repeated forward hops (regular)	0.673	5.3 ± 1.2	0.7 ± 0.2	77.0 ± 11.2
	1-Leg countermovement hop	0.711	4.7 ± 1.0	4.5 ± 1.3	57.4 ± 15.3
	Run-and-cut	0.725	5.5 ± 1.2	0.7 ± 0.2	96.2 ± 18.3
	1-Leg decline squat	0.747	5.2 ± 0.6	6.8 ± 2.0	12.3 ± 2.4

[a] Loading index is the weighted sum of scaled and normalized peak loading, loading impulse, and loading rate.
BW, body weight; 1-Leg, single-leg; 2-Leg, double-leg; full, complete range of motion; regular, self-selected speed; fast, faster speed.

knowledge, this is the first study to establish a comprehensive biomechanical set of patellar tendon loading data based on a battery of exercises commonly prescribed by clinicians. By defining a patellar tendon loading index that incorporates three components of patellar tendon loading, this study allows clinicians to rank exercises based on loading peak, loading impulse, and loading rate. These loading guidelines will be especially useful for clinicians to better prescribe progressive

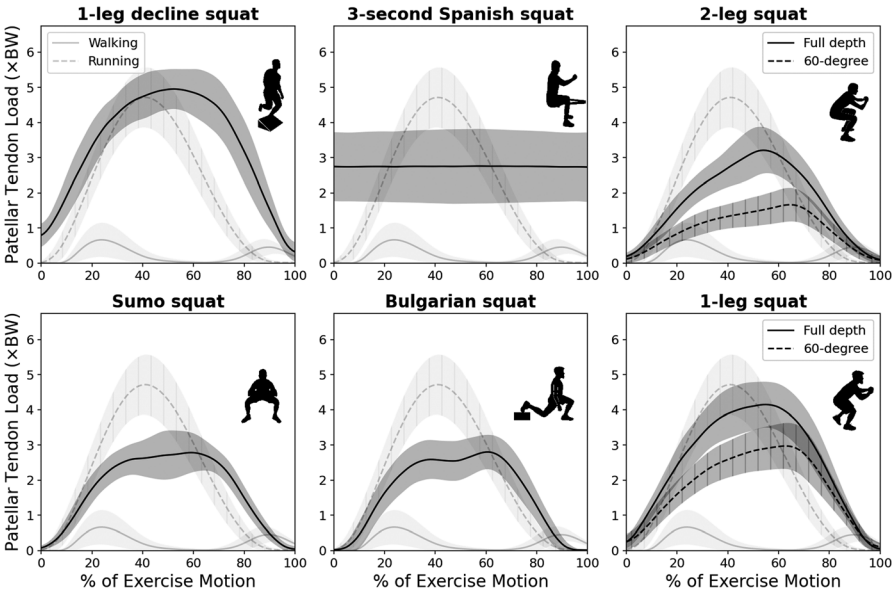

FIGURE 3—Patellar tendon loading during different variations of the squat exercise. Group averages are represented as solid or dashed lines, and the SD is represented as shades. Patellar tendon loading during self-selected pace walking and running are graphed in *light gray solid* and *dashed lines*, respectively. The *x*-axis indicates normalized exercise time, and the *y*-axis indicates patellar tendon load normalized against body weight. BW, body weight; 1-Leg, single-leg; 2-Leg, double-leg.

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therapeutic loading for patients with knee disorders like patellar tendinopathy and ACL reconstruction using bone–patellar tendon–bone grafts. A modifiable worksheet (Supplemental Digital Content 2, http://links.lww.com/MSS/C927) is also provided so that the clinician can use it to automatically rerank and recategorize exercises according to the relative importance assigned to loading peak, impulse, and rate for their specific patient. In the beginning stages of rehabilitation, the clinician might want to rank the exercises based on loading peak and loading impulse, aiming to progressively increase the tendon's load tolerance. In the advanced stages, however, to adequately prepare the patient for return to sport, the clinician might want to rerank the exercises based on how fast the load is applied to the tendon, in which case the loading rate would have a greater importance.

The single-leg decline squat is the most common exercise used at the beginning of rehabilitation to treat patellar tendinopathy patients (8–14). Some authors even consider the eccentric training, using the single-leg decline squat, the gold standard first-line management approach for patellar tendinopathy, mostly due to this exercise having the largest body of evidence supporting its effects (41,42). However, several authors have shown that improvements in pain and function in tendinopathy patients occur with progressive loading protocols regardless of contraction type (11,15,18,19), and there is no convincing evidence that isolated eccentric exercises are superior to other loading therapies (15–17). Interestingly, in the present study, the single-leg decline squat showed the highest loading index, with very high loading peak (5.2 body weights) and impulse (6.8 body weights·second). This indicates that using the single-leg decline squat at the beginning of patellar tendinopathy rehabilitation is not the best approach, considering that the treatment premise is building a progressive load tolerance to the tendon. This may be one of the reasons why interventions based exclusively on the eccentric single-leg decline squat exercise have lower patient satisfaction in comparison with other exercise interventions (11,15) and less than optimal results in the long-term, with 45% of patients not being able to return to sport (10). Also, the use of the single-leg decline squat as a prevention exercise during season had the opposite effect, with asymptomatic athletes developing patellar tendinopathy after the intervention (43). Given the results of the current study, it is possible that this is because the single-leg decline squat generates a large amount of force in the patellar tendon, which, in addition to the in-season loads generated in sports participation, could cause overuse instead of a therapeutic load.

In the present study, the Spanish squat exercise had a moderate loading index (0.563) with this value mostly explained by loading impulse (cumulative load in the tendon). Recently, the Spanish squat has emerged as an exercise of choice for the treatment of patellar tendinopathy patients (32,34,36,44). The Spanish squat is being recommended in the initial stages of rehabilitation and is considered a particularly useful exercise for the treatment of athletes on the road, when they have no access to gym equipment (44). However, in the present study, several other exercises, including other variations of the squat (i.e., Bulgarian squat, sumo squat, double and single-leg 60° squats), showed lower loading indexes than the Spanish squat. In this context, starting the rehabilitation with these exercises, later progressing to the Spanish squat at an intermediate stage, might be a better approach for the treatment of patellar tendinopathy patients. After ACL reconstruction using bone–patellar tendon–bone grafts, the clinician should keep in mind that the Spanish squat not only produces a substantial amount of load in the healing patellar tendon but also generates a significant anterior tibial translation force, which is contraindicated in the initial stages of graft healing.

The results of patellar tendon loading in the present study are similar to those of previous studies (21,45). The analysis of the different variations of the squat exercise in the present study showed that single-leg squats did not double peak loading compared with double-leg squats. However, 60° squats resulted in approximately half the peak loading when compared with full squats. This finding is in accordance with the results of a recent study that found that deeper squats (>60°) generate significantly more muscle activation of the rectus femoris, vastus medialis, and vastus lateralis (46). In this context, to progressively load the patellar tendon in rehabilitation, the clinician should consider progressing from double-leg squats to single-leg squats before progressing from 60° squats to full squats. The same rationale might be useful in the early stages after ACL reconstruction using bone–patellar tendon–bone grafts, when graft site healing is still occurring, and excessive load could cause overuse in the patellar tendon.

Recently, jump–landing trainings have been included in the rehabilitation protocols of patellar tendinopathy patients (14,33). This approach comes from the observation that athletes with patellar tendinopathy have stiffer landing mechanics, with smaller range of motion in the lower limb joints after foot contact (47). More recent reviews have also highlighted that athletes with patellar tendinopathy and those with ACL reconstruction have an abnormal landing mechanics (48–50). The inclusion of commonly prescribed jumps and jump–landing exercises in this study was based on the need to understand how much load the patellar tendon undergoes during these exercises in relation to other rehabilitation exercises. The results showed that most jumping and landing exercises generated moderate loads in the patellar tendon, being categorized as tier 2 exercises. The only exceptions were the single-leg maximal forward hop, the single-leg countermovement jump, and the single-leg repeated forward hops, which generated higher loads, being categorized as tier 3 exercises. This indicates that jumping and landing exercises could potentially be introduced at an intermediate stage in the rehabilitation of patellar tendinopathy patients, with the exception of the higher loading index exercises.

The clinician should also keep in mind that the patellar tendon force during landing is influenced by trunk position (24). Verbal instructions to encourage patients to increase trunk flexion and land softly have been shown to improve the load transmission between the three major lower limb joints in patellar tendinopathy patients, also resulting in significant improvements in terms of pain and function (14,33). In this context, trunk position during landing exercises is a variable that can be modulated to decrease patellar tendon load in the initial stages of rehabilitation and to increase the load in the advanced stages. Jumping and landing exercises are also frequently prescribed in the rehabilitation after ACL

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reconstruction (51,52), also with the use of verbal instructions to improve jump–landing mechanics (53,54). The findings of the current study may help clinicians establish an order of progression in the jump and landing training. The tiering system also provides clinicians with more freedom to tailor rehabilitation protocols based on patient's specific needs and preferences while meeting tendon loading goals in a clear, well-established progression (14).

This study has limitations that need to be acknowledged. Only young healthy males and females participated in this study; therefore, the generalization of these results to other populations, such as patellar tendinopathy patients, should be done with caution. Another limitation is the fact that by defining a constrained knee with flexion-extension only, the model did not consider the effects of frontal or transverse plane movements on patellar tendon loading. This decision was made based on the fact that past studies reported that patellar tendon forces primarily act in the sagittal plane (21,22). However, because frontal and transverse movements may affect patellar tendon forces, especially during single-limb exercises (55), future research should include those aspects in the model. A third limitation is the fact that the model used net knee moment to estimate the quadriceps loading required for knee extension, which neglects the potential effects of coactivation and force production by the hamstrings. Further studies should examine the activation of the quadriceps and hamstring muscles to verify whether patellar tendon loading is influenced by muscle coactivations during these exercises. The order of the exercises was not random, so fatigue may have influenced the movement patterns and consequently the tendon forces of the last exercises of each module. Lastly, the loading index was calculated as a weighted summation of peak load, loading impulse, and loading rate (28) based on the fact that the tendon is a viscoelastic tissue (56). Although the authors' intuition was used to scale these three loading parameters, changing the scaling factors of each of these parameters may affect the exercise ranking. To minimize this limitation, a spreadsheet is provided so that clinicians can recalculate the exercise rankings based on their interpretation about the importance of each loading parameter. Future studies should investigate the effects of interventions using the loading tiers exercise progression in the rehabilitation of patients with knee disorders such as patellar tendinopathy and after ACL reconstruction.

CONCLUSIONS

An exercise ranking that progressively increases patellar tendon loading across a diverse set of exercises routinely used in the treatment of patients with knee disorders was established. Three loading tiers were established based not only on loading peak but also on loading impulse and loading rate. This knowledge may help clinicians better understand the loads that act on the tendon during different exercises and may help with the tailoring of interventions for the rehabilitation of patients with patellar tendon disorders and after ACL reconstruction with a bone–patellar tendon–bone autograft.

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